1. Title

PolicyGPT: Government Policy & Public Scheme Intelligence Platform

2. Objective

Build a full-stack web application using Angular and FastAPI that enables citizens, government officials, researchers, students, and organizations to access, search, analyze, and understand government policies, public welfare schemes, regulations, notifications, circulars, and official documents through a centralized digital platform.

The platform provides intelligent policy search, document categorization, eligibility checking, scheme comparison, policy tracking, notification management, and analytics dashboards to simplify access to government information.

The system is designed to improve transparency, increase awareness of government schemes, and help users quickly discover relevant policies and benefits.

This solution can be used by central and state government departments, educational institutions, NGOs, researchers, businesses, and citizens.

Outcomes

(i) Developed and deployed a full-stack Government Policy Intelligence Platform.

(ii) Implemented secure user authentication and role-based access control.

(iii) Built centralized policy and public scheme management modules.

(iv) Implemented intelligent policy search and filtering mechanisms.

(v) Developed scheme eligibility checking and comparison features.

(vi) Built policy tracking and notification systems.

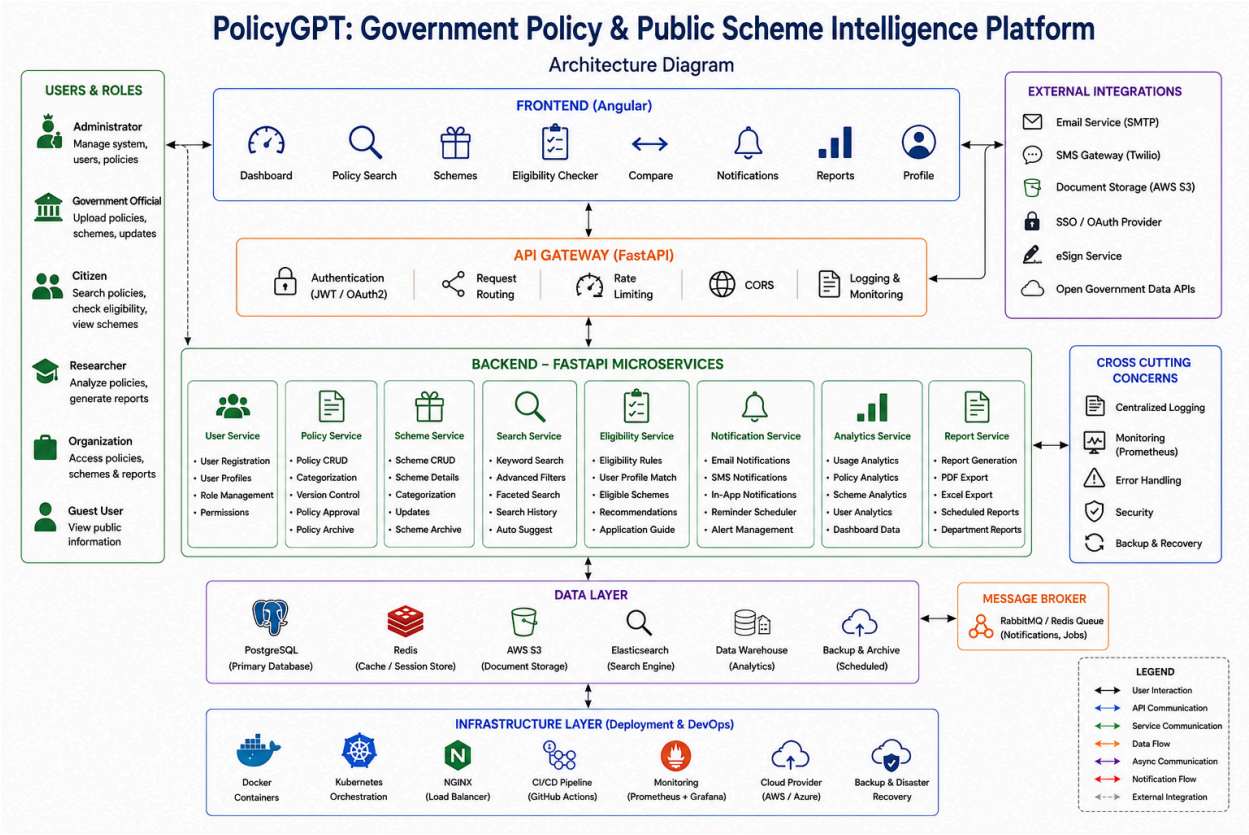
(vii) Developed dashboards for policy analytics and usage statistics.

(viii) Generated reports for policies, schemes, and citizen engagement.

(ix) Deployed the application using Docker and cloud platforms.

(x) Designed responsive UI wireframes and government service workflows before implementation.

3. Architecture Components



4. Modules to be Implemented

1. User Authentication & Role Management

Features

(i) User Registration

(ii) Secure Login

(iii) JWT Authentication

(iv) Password Reset

(v) Profile Management

(vi) Role-Based Access Control

Roles

Administrator

Government Official

Citizen

Researcher

Organization

Guest User

2. Policy Management Module

Features

(i) Policy Upload

(ii) Policy Categorization

(iii) Policy Editing

(iv) Policy Approval

(v) Policy Search

(vi) Policy Archive

Policy Categories

Education

Healthcare

Agriculture

Employment

Finance

Women & Child Welfare

Housing

Environment

Digital Governance

Infrastructure

3. Public Scheme Management Module

Features

(i) Scheme Registration

(ii) Scheme Details Management

(iii) Scheme Categorization

(iv) Scheme Eligibility Rules

(v) Scheme Updates

(vi) Scheme Archive

Scheme Categories

Scholarships

Farmer Welfare

Healthcare

Housing

Business Support

Women Empowerment

Senior Citizen Welfare

Student Schemes

Employment Programs

Social Security

4. Policy Search Module

Features

(i) Keyword Search

(ii) Advanced Filters

(iii) Category-wise Search

(iv) State-wise Search

(v) Ministry-wise Search

(vi) Department-wise Search

Search Filters

Policy Name

Scheme Name

Department

State

Sector

Publication Date

Status

5. Eligibility Checker Module

Features

(i) Eligibility Form

(ii) User Profile Analysis

(iii) Scheme Matching

(iv) Eligibility Summary

(v) Recommended Schemes

(vi) Application Guidance

Eligibility Parameters

Age

Gender

Income

Occupation

Education

Location

Social Category

Disability Status

6. Policy Comparison Module

Features

(i) Compare Policies

(ii) Compare Government Schemes

(iii) Compare Benefits

(iv) Compare Eligibility

(v) Compare Departments

(vi) Compare Application Process

7. Notification Module

Features

(i) New Policy Alerts

(ii) Scheme Update Notifications

(iii) Deadline Reminders

(iv) Application Notifications

(v) Email Notifications

(vi) SMS Notifications

(vii) In-App Notifications

8. Dashboard & Analytics Module

Citizen Dashboard

(i) Saved Policies

(ii) Eligible Schemes

(iii) Recent Notifications

(iv) Search History

(v) Application Status

Government Dashboard

(i) Policy Statistics

(ii) Scheme Usage Analytics

(iii) User Activity

(iv) Department Reports

(v) Notification Statistics

Admin Dashboard

(i) User Management

(ii) Policy Management

(iii) Analytics

(iv) Reports

(v) Audit Logs

9. Reports & Export Module

Features

(i) Policy Reports

(ii) Scheme Reports

(iii) User Activity Reports

(iv) Department Reports

(v) Analytics Reports

(vi) PDF Export

(vii) Excel Export

10. Feedback & Support Module

Features

(i) Citizen Feedback

(ii) Issue Reporting

(iii) Help Desk

(iv) FAQ Management

(v) Query Resolution

(vi) Contact Support

11. Final Integration, Testing & Deployment

5. Milestone 1: Week 1 & 2 — Requirements, UI Design, Database Design & Backend Setup

Tasks

(i) Define project objectives.

(ii) Gather functional and non-functional requirements.

(iii) Design UI wireframes.

(iv) Create user flow diagrams.

(v) Prepare dashboard layouts.

(vi) Design responsive screens using Figma.

(vii) Design database schema.

Tables

users

policies

schemes

eligibility_rules

notifications

feedback

reports

audit_logs

search_history

(viii) Initialize FastAPI project.

(ix) Configure PostgreSQL.

(x) Implement JWT Authentication.

(xi) Create Angular frontend.

(xii) Configure Angular Material.

UI Wireframes

Login Page

Registration Page

Citizen Dashboard

Government Dashboard

Policy Search Page

Policy Details Page

Scheme Details Page

Eligibility Checker

Reports Dashboard

Notification Screen

Outcomes

(i) UI wireframes completed.

(ii) User flows finalized.

(iii) Database schema completed.

(iv) FastAPI backend initialized.

(v) Angular frontend initialized.

(vi) Authentication implemented.

6. Milestone 2: Week 3 & 4 — Policy & Scheme Management

Tasks

(i) Develop Policy Management module.

(ii) Build Scheme Management module.

(iii) Implement Search functionality.

(iv) Create Policy Approval workflow.

(v) Build Eligibility Checker.

(vi) Develop Policy Comparison module.

Outcomes

(i) Policy repository operational.

(ii) Scheme management completed.

(iii) Search functionality operational.

(iv) Eligibility checker functional.

7. Milestone 3: Week 5 & 6 — Analytics, Notifications & Reporting

Tasks

(i) Develop Analytics Dashboard.

(ii) Build Notification System.

(iii) Generate Reports.

(iv) Implement Feedback module.

(v) Develop Department Analytics.

(vi) Build Usage Statistics Dashboard.

Outcomes

(i) Analytics dashboard completed.

(ii) Notification system operational.

(iii) Reports generated successfully.

(iv) Feedback management completed.

8. Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

(i) Perform application testing.

(ii) Fix identified issues.

(iii) Deploy frontend and backend.

(iv) Configure Docker.

(v) Prepare documentation.

(vi) Demonstrate complete application.

Outcomes

(i) Production-ready application.

(ii) Deployment completed.

(iii) Documentation completed.

(iv) Successful project demonstration.

9. Evaluation Criteria

Milestone 1

(i) FastAPI project setup completed.

(ii) Angular application initialized.

(iii) Database schema finalized.

(iv) Authentication implemented.

(v) UI wireframes completed.

Milestone 2

(i) Policy repository operational.

(ii) Scheme management functional.

(iii) Search functionality completed.

(iv) Eligibility checker operational.

Milestone 3

(i) Analytics dashboard completed.

(ii) Notification system operational.

(iii) Reports generated successfully.

(iv) Feedback module functional.

Milestone 4

(i) Application deployed successfully.

(ii) Testing completed.

(iii) Documentation prepared.

(iv) Complete workflow demonstrated.

10. Tools & Tech Stack

Programming Language

Backend

Python

FastAPI

Frontend

TypeScript

Angular

Database

PostgreSQL

Redis

Backend Framework

FastAPI

SQLAlchemy

Pydantic

Alembic

Uvicorn

Celery

Frontend Framework

Angular

Angular Material

Bootstrap

RxJS

Chart.js

Authentication

JWT Authentication

OAuth2

Role-Based Access Control

Notifications

SMTP Email

Twilio SMS

Firebase Cloud Messaging

Cloud & DevOps

Docker

Docker Compose

GitHub Actions

AWS

Azure

Nginx

Development Tools

VS Code

Git

GitHub

Postman

Figma